

Clinical Commissioning Group Analysis Report (NHS Trusts)

Stakeholder Report

Summary

This report delves into the data provided by the Department of Health and Social Care for the year 2021. It is specifically centered on the populations of each **Clinical Commissioning Group (CCG)** and the numbers of stroke and heart disease mortalities. The task was to identify the top 5 CCGs with the highest stroke and heart disease mortality rates. I was also required to recommend a method(s) to predict the number of stroke mortalities for the following year, 2022. NHS Blackpool had the highest mortality rate, shortly followed by CCGs scattered across the country.

CCG with **highest** mortality rate:

NHS Blackpool

0.261

What are the top 5 CCGs with the highest mortality rates?

CCG Name	Mortality Rate
NHS Blackpool CCG	0.261
NHS South Sefton CCG	0.239
NHS North East Lincolnshire CCG	0.238
NHS North Cumbria CCG	0.234
NHS Fylde and Wyke CCG	0.230

Mortality rates are close in values, having a difference of 0.031% across 5 CCGs. Upon plotting a distribution to see how mortality rates are distributed amongst CCGs, the majority of cases fall within a mortality rate of 0.125% to 0.20%, with the most typical rate around 0.15%. A small number of cases exhibit significantly higher rates above 0.225%, which may warrant further investigation.

Key Findings:

Three of the top 5 CCGs are within proximity to each other geographically. Blackpool, South Sefton, and Fylde and Wylde are all situated on the North West of England. Upon further research, key drivers of their highest mortality rates are:

- ⇒ Deprived socioeconomic levels
- ⇒ Higher prevalence of smoking
- ⇒ Higher prevalence of unmanaged High blood pressure
- ⇒ Poorer diets

All these factors contribute to an increased stroke and heart disease risk, which consequently results in higher rates of mortalities from those risks.

Machine Learning to predict?

To predict rates for upcoming years, we would need access to years of historical data. This data can be used to train a Poisson regression model that can predict mortality rates for 2022. To validate the model, I would apply **k-fold cross-validation** across CCGs and assess performance using MAE and RMSE against known 2021 figures before projecting to 2022.

Due to the small capacity of the sample size, machine learning is not the most viable option. If larger datasets were obtained, ML would work well.

Recommendations:

- Investigate methods in increasing awareness of the public on the higher mortality rates
- Collaborate with local councilors, public health officials, NHS boards, etc. to build campaigns that provide care and support to those who need it
- Use a Poisson regression model to predict data for the years to come.